

The Neuron

Note: In OneNote, set this worksheet as background. Use "Export" in the simulation and insert the image in the marked space.

The neuron is a specialised cell of the nervous system. It receives information, processes it and passes it on. The dendrites increase the surface area of the cell and receive signals from other neurons. The cell body, also called the soma, contains the nucleus and many organelles. Metabolic processes are controlled here and incoming information is brought together. At the axon hillock begins the long extension of the neuron, the axon. The axon hillock is especially important because it is where it is decided whether the incoming signals will produce a transmitted electrical impulse. If the excitation is strong enough, a signal is triggered. It spreads along the axon to the terminal buttons. Many axons have a myelin sheath. This insulating covering protects the axon and ensures that conduction is much faster. At the terminal buttons, electrical conduction does not simply end; instead, the information is passed on to the next cell. This happens at the synapse. There, messenger substances are released, travel through the synaptic cleft to the next cell and trigger or influence a new signal there.

In the neuron, several tasks can therefore be clearly distinguished: receiving information, processing it in the cell body, conducting it in the axon and transmitting it at the synapse.

Task 1: Label the neuron

Export the labelled figure from the "Neuron" mode and insert it below.

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Task 2: Matching structure and function

Complete the matching task, export the correct solution and insert it below.

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Task 3: Signal transmission along the neuron

Describe the signal path from dendrites through soma and axon to the terminal buttons. Use technical terms.

This image shows a single sheet of white paper with horizontal blue ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.